

Ratio and Regression Estimation

SurvMeth/Surv 625: Applied Sampling

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Review: Stratified sampling

- 1 Identify stratifying variables correlated with the measure(s) of interest
- 2 Choose “cuts” on the stratifying variables and divide the population into strata
- 3 S_h , $deff$, and S^2 are known, at least approximately, and known that fpc's within strata can be ignored
- 4 Compute an stratified sample size $n = n_{SRS} * deff$
- 5 Determine an allocation for the desired n
- 6 Adjust n based on expected $deff$ & allocate
- 7 Select sample and compute estimates taking the stratified sample selection into account

Model-based theory

- In randomization theory, or design-based sampling, the **sampling design determines how sampling variability is estimated**.
- In model-based sampling, the **model determines how variability is estimated**, and the sampling design is irrelevant—as long as the model holds, you could choose any n units you want to from the population.
- The discrepancy is due to the different definitions of variance:
 - In design-based sampling, the variance is the average squared deviation of the estimate from its expected value, averaged over all samples that could be **obtained using a given design**.
 - If we are using a model, the variance is again the average squared deviation of the estimate from its expected value, but here the average is over all possible samples that could be **generated from the population model**.

Model-based theory cont.

- Design-based inferences about finite population quantities using ratio or regression estimation are correct *even if the model does not fit the data well*.
- Model-assisted estimators: A model motivates the form of the estimator, but inference depends on the sampling design.
- If we adopt a model consistent with the reasons we would adopt a certain sampling scheme or method of estimation, the point estimators would be similar. The model-based variance, though, usually differs from the variance from the randomization theory.

Ratio and regression estimation

- Ratio estimation is used to improve the precision of estimates by incorporating auxiliary information correlated with the variable of interest.
- Suppose we want to estimate the total or mean of a population characteristic Y . We have an auxiliary variable X for which the total is known, or its sample mean can be accurately estimated.
- The ratio estimator uses both Y and X to improve estimation.
- Define the ratio $B = \frac{t_y}{t_x} = \frac{\bar{Y}}{\bar{X}}$.
- Ratio and regression estimation both use the correlation of X and Y . The population correlation coefficient of X and Y is

$$R = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{(N-1)S_x S_y}$$

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- Ratio estimator $\hat{B} = \frac{\hat{t}_y}{\hat{t}_x} = \frac{\bar{y}}{\bar{x}}$, and $\hat{y}_r = \hat{B}\bar{X}$

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- Adjust estimates from the sample to reflect population totals (poststratification)

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- Adjust for nonresponse (discussed later)

Examples

- Using the total number of registered births as auxiliary information for estimating the total population in France (Laplace, 1814)
- Using field acreage to estimate field yield of grains
- Using farm acreage in 1987 to estimate farm acreage in 1992
- Using family income to estimate wealth
- Using the number of employees to estimate the amount spent on health insurance in a business

Poststratification: Example

- An SRS of 400 students taken from a school with 4000 students: 240 women and 160 men
- With 84 of the sampled women and 40 of the sampled men planning to follow careers in academia
- Question: How many students planning to work in academia?
- ① SRS: $\frac{4000}{400} * 124 = 1240$
- ② If we know that the school has 2700 women and 1300 men, another estimate is $\frac{2700}{240} * 84 + \frac{1300}{160} * 40 = 1270$
- We can treat this as a ratio estimation by gender:
$$\frac{84}{240} * 2700 + \frac{40}{160} * 1300 = 1270$$
 - The sample has 60% women, but the population has 67.5%.
 - We adjust the estimated total by the sexual decomposition discrepancy:

Poststratification

Ratio estimators are usually biased

- If we calculate $\hat{y}_r = \bar{y} \frac{\bar{X}}{\bar{x}}$ for all possible SRSs, their average value will be close to but usually not be equal $\bar{Y}$ exactly.

$$Bias(\hat{y}_r) = E(\hat{y}_r - \bar{Y}) = -Cov(\hat{B}, \bar{x})$$

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- We have

$$\frac{|Bias(\hat{y}_r)|}{\sqrt{Var(\hat{y}_r)}} = \frac{|Cov(\hat{B}, \bar{x})|}{\bar{X} \sqrt{Var(\hat{B})}} \leq \frac{\sqrt{Var(\bar{x})}}{\bar{X}} = CV(\bar{x})$$

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- The absolute value of the bias is small relative to the standard deviation if $CV(\bar{x})$ is small.
- A small $CV(\bar{x})$ means that $\bar{x}$ is stable from sample to sample

Approximation for the bias

- Linearization approach to approximating variances

$$\hat{\bar{y}}_r - \bar{Y} = (\bar{y} - B\bar{x})\left(1 - \frac{\bar{x} - \bar{X}}{\bar{x}}\right)$$

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$$Bias(\hat{y}_r) = E(\hat{y}_r - \bar{Y}) \approx (1 - \frac{n}{N}) \frac{1}{n\bar{X}(BS_x^2 - RS_xS_y)} \quad (1)$$

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 - n/N is large
 - $\bar{X}$ is large
 - S_x is small
 - R is close to 1

Ratio estimation with weights

- With the sampling weight $w_i = 1/\pi_i$, $\bar{y}_w = \frac{\sum_{i \in s} w_i y_i}{\sum_{i \in s} w_i}$

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- Define $g_i = \frac{t_x}{\hat{t}_x}$, then $\hat{t}_{yr} = \sum_{i \in s} w_i g_i y_i$ as a weighted sum of the observations, with new weights

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- Define $g_i = \frac{t_x}{\hat{t}_x}$, then $\hat{t}_{yr} = \sum_{i \in s} w_i g_i y_i$ as a weighted sum of the observations, with new weights
- The weight adjustments g_i calibrates the estimates based on x :
$$\sum_{i \in s} w_i g_i x_i = t_x.$$

Mean squared error (MSE)

- We have

$$E[(\hat{y}_r - \bar{Y})^2] = E[\{(\bar{y} - B\bar{x})(1 - \frac{\bar{x} - \bar{X}}{\bar{x}})\}^2] \approx E[(\bar{y} - B\bar{x})^2]$$

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- The **variance** and **MSE** are both approximated by

$$MSE(\hat{y}_r) = E[(\bar{y} - B\bar{x})^2] = (1/n - 1/N)(S_y^2 - 2BR S_x + B^2 S_x^2)$$

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- The MSE is small if n is large, n/N is large, the deviations $d_i = y_i - Bx_i$ is small, and R is close to 1.

Variance estimator based on regression residuals

- Since $E[(\bar{y} - B\bar{x})^2] = Var(\bar{d})$, $\bar{d}_U = 0$, define a new variable $e_i = y_i - \hat{B}x_i$ as the i -th residual from fitting the line $y = \hat{B}x$.

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- Estimate $var(\hat{y}_r) = (1 - \frac{n}{N})(\frac{\bar{X}}{\bar{x}})^2 \frac{s_e^2}{n}$ with $s_e^2 = \frac{1}{n-1} \sum_{i \in S} e_i^2$

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- With weights $g_i = \frac{\bar{X}}{\bar{x}}$ and $u_i = g_i e_i$, for an SRS,
$$var(\bar{u}) = (1 - \frac{n}{N}) \frac{1}{n(n-1)} \sum_{i \in S} (u_i - \bar{u})^2 = (1 - \frac{n}{N})(\frac{\bar{X}}{\bar{x}})^2 \frac{s_e^2}{n} = var(\hat{y}_r)$$

Variance estimator based on regression residuals

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- Estimate $\text{var}(\hat{y}_r) = (1 - \frac{n}{N})(\frac{\bar{X}}{\bar{x}})^2 \frac{s_e^2}{n}$ with $s_e^2 = \frac{1}{n-1} \sum_{i \in S} e_i^2$
- With weights $g_i = \frac{\bar{X}}{\bar{x}}$ and $u_i = g_i e_i$, for an SRS,
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- Similarly

$$\text{var}(\hat{B}) = (1 - \frac{n}{N}) \frac{s_e^2}{n\bar{x}^2} \quad (2)$$

R code: Examples 4.2 in Lohr

```
agsrs$sampwt <- rep(3078/n,n)
agdsrs <- svydesign(id = ~1, weights=~sampwt, fpc=rep(3078,300), data = agsrs)
# estimate the ratio acres92/acres87
sratio<-svyratio(numerator = ~acres92, denominator = ~acres87,design = agdsrs); sratio
```

```
Ratio estimator: svyratio.survey.design2(numerator = ~acres92, denominator = ~acres87,
design = agdsrs)
```

```
Ratios=
      acres87
acres92 0.9865652
SEs=
```

```
      acres87
acres92 0.005750473
```

```
confint(sratio, df=degf(agdsrs))
```

```

      2.5 %    97.5 %
acres92/acres87 0.9752487 0.9978818
# provide the population total of x
xpopttotal <- 964470625
# Ratio estimate of population total
predict(sratio,total=xpopttotal)
```

```
$total
      acres87
acres92 951513191
```

```
$se
      acres87
acres92 5546162
```

```
# Ratio estimate of population mean
predict(sratio,total=xpopttotal/3078)
```

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Regression estimation in SRS

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- General regression model $y = B_0 + B_1x$ with ordinary least squares regression coefficient estimate

$$\hat{B}_1 = \frac{\sum_{i \in s} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i \in s} (x_i - \bar{x})^2} = \frac{rs_y}{s_x}$$

$$\hat{B}_0 = \bar{y} - \hat{B}_1 \bar{x}$$

Regression estimation in SRS

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$$\hat{B}_0 = \bar{y} - \hat{B}_1 \bar{x}$$

- If we know $\bar{X}$, we can obtain the regression estimator $\hat{y}_{reg} = \hat{B}_0 + \hat{B}_1 \bar{X} = \bar{y} + \hat{B}_1 (\bar{X} - \bar{x})$

Regression estimator is biased

- Let population slope $B_1 = \frac{RS_y}{S_x}$, we have

$$E(\hat{y}_{reg} - \bar{Y}) = -Cov(\hat{B}_1, \bar{x})$$

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$$E(\hat{y}_{reg} - \bar{Y}) = -Cov(\hat{B}_1, \bar{x})$$
- For large SRSs the MSE for regression estimation is similar to the variance. Let $d_i = y_i - [\bar{Y} + B_1(x_i - \bar{X})]$, then

$$MSE(\hat{y}_{reg}) = E[\{\bar{y}\bar{Y} + B_1(x_i - \bar{X})\}^2] \approx V(\bar{d}) = (1 - \frac{n}{N}) \frac{S_d^2}{n}$$

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- We can estimate S_d^2 by using residuals $e_i = y_i - (\hat{B}_0 + \hat{B}_1 x_i)$ and $s_e^2 = \sum_i e_i^2 / (n - 2)$

$$SE(\hat{y}_{reg}) = \sqrt{(1 - \frac{n}{N}) \frac{1}{n} s_y^2 (1 - r^2)} \quad (3)$$

R code: Lohr Example 4.7

```
data(deadtrees); #head(deadtrees) nrow(deadtrees) # 25
# Fit with survey regression
dtree<- svydesign(id = ~1, weight=rep(4,25), fpc=rep(100,25), data = deadtrees)
myfit1 <- svyglm(field~photo, design=dtree)
summary(myfit1) # displays regression coefficients
```

Call:

```
svyglm(formula = field ~ photo, design = dtree)
```

Survey design:

```
svydesign(id = ~1, weight = rep(4, 25), fpc = rep(100, 25), data = deadtrees)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.0593	1.3930	3.632	0.0014 **
photo	0.6133	0.1259	4.870	6.44e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 5.548341)

Number of Fisher Scoring iterations: 2

```
confint(myfit1,df=23) # df = 25-2
```

	2.5 %	97.5 %
(Intercept)	2.1777362	7.940848
photo	0.3527717	0.873777

R code: Lohr Example 4.7 cont.

```
# Regression estimate of population mean field trees
newdata <- data.frame(photo=11.3)
predict(myfit1, newdata)
```

```
      link      SE
1 11.989 0.418
confint(predict(myfit1, newdata),df=23)
```

```
      2.5 %   97.5 %
1 11.12455 12.85404
# Estimate total field tree, add population size in total= argument
newdata2 <- data.frame(photo=1130)
predict(myfit1, newdata2, total=100)
```

```
      link      SE
1 1198.9 41.802
confint(predict(myfit1, newdata2,total=100),df=23)
```

```
      2.5 %   97.5 %
1 1112.455 1285.404
```

Subdomain estimation

- Often we want separate estimates for subpopulations; the subpopulations are called domains or subdomains.
- The number of persons in an SRS who fall into each domain n_d is a random variable, so estimating domain means is a special case of ratio estimation.
- Let $x_i = 1$ and $u_i = x_i y_i = y_i$ if $i \in s_d$ and $x_i = u_i = 0$ otherwise
- Domain mean $\bar{y}_d = \frac{\hat{t}_u}{\hat{t}_x} = \frac{\bar{u}}{\bar{x}} = \hat{B}$

Subdomain estimation cont.

- Using the variance of $\hat{B}$ in Equation (2),

$$SE(\bar{y}_d) = \sqrt{\left(1 - \frac{n}{N}\right) \frac{n}{n_d^2} \frac{(n_d - 1)s_{yd}^2}{n - 1}},$$

where $s_{yd}^2 = \sum_{i \in s_d} (y_i - \bar{y}_d)^2 / (n_d - 1)$

Subdomain estimation cont.

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where $s_{yd}^2 = \sum_{i \in s_d} (y_i - \bar{y}_d)^2 / (n_d - 1)$

- If $E(n_d)$ is large, $SE(\bar{y}_d) \approx \sqrt{\left(1 - \frac{n}{N}\right) \frac{s_{yd}^2}{n_d}}$

Subdomain estimation cont.

- Using the variance of $\hat{B}$ in Equation (2),

$$SE(\bar{y}_d) = \sqrt{\left(1 - \frac{n}{N}\right) \frac{n}{n_d^2} \frac{(n_d - 1)s_{yd}^2}{n - 1}},$$

where $s_{yd}^2 = \sum_{i \in s_d} (y_i - \bar{y}_d)^2 / (n_d - 1)$

- If $E(n_d)$ is large, $SE(\bar{y}_d) \approx \sqrt{\left(1 - \frac{n}{N}\right) \frac{s_{yd}^2}{n_d}}$
- The results for now only apply to SRSs, and approximations depends on having a sufficiently large sample so that $E(n_d)$ is large

Subdomain estimation cont.

- Using the variance of $\hat{B}$ in Equation (2),

$$SE(\bar{y}_d) = \sqrt{\left(1 - \frac{n}{N}\right) \frac{n}{n_d^2} \frac{(n_d - 1)s_{yd}^2}{n - 1}},$$

where $s_{yd}^2 = \sum_{i \in s_d} (y_i - \bar{y}_d)^2 / (n_d - 1)$

- If $E(n_d)$ is large, $SE(\bar{y}_d) \approx \sqrt{\left(1 - \frac{n}{N}\right) \frac{s_{yd}^2}{n_d}}$
- The results for now only apply to SRSs, and approximations depends on having a sufficiently large sample so that $E(n_d)$ is large
- More generally, we use small area estimation approaches that rely on models

R code: Lohr Example 4.8

```
agsrsnew<-agsrs
agsrsnew$farmcat<-rep("large",n)
agsrsnew$farmcat[agsrsnew$farms92 < 600] <- "small" #head(agsrsnew)
dsrsnew <- svydesign(id = ~1, weights=~sampwt, fpc=rep(3078,300), data=agsrsnew)
# domain estimation for large farmcat with subset statement
dsub1<-subset(dsrsnew,farmcat=='large') # design info for domain large farmcat
smean1<-svymean(~acres92,design=dsub1); smean1
```

```
      mean      SE
acres92 316566 21553
df1<-sum(agsrsnew$farmcat=='large')-1; df1 #calculate domain df if desired
```

```
[1] 128
confint(smean1, level=.95,df=df1) # CI
```

```
      2.5 %    97.5 %
acres92 273918.9 359212.4
# use svyby function
bothmeans<-svyby(~acres92,by=~factor(farmcat),design=dsrsnew,svymean); bothmeans
```

```
      factor(farmcat) acres92      se
large      large 316565.7 21553.21
small      small 283813.7 28852.24
confint(bothmeans,level=.95)
```

```
      2.5 %    97.5 %
large 274322.1 358809.2
small 227264.4 340363.1
```

Poststratification

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- If n_h is reasonably large, we can use an approximate variance estimator $var(\bar{y}_{post}) \approx (1 - \frac{n}{N}) \sum N_h / N s_h^2 / n$
- Difference between stratification (design) and poststratification (estimation): n_h fixed or random?

R code: Lohr Example 4.9

```
data(agsrs)
dsrs <- svydesign(id = -1, weights=rep(3078/300,300), fpc=rep(3078,300),data = agsrs)
# Create a data frame that gives the population totals for the poststrata
pop.region <- data.frame(region=c("NC","NE","S","W"), Freq=c(1054,220,1382,422))
# create design information with poststratification
dsrsp<-postStratify(dsrs, ~region, pop.region); summary(dsrsp)
```

```
Independent Sampling design
postStratify(dsrs, ~region, pop.region)
Probabilities:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.09242 0.09407 0.09407 0.09771 0.10152 0.10909
Population size (PSUs): 3078
Data variables:
 [1] "county"  "state"    "acres92"  "acres87"  "acres82"  "farms92"
 [7] "farms87"  "farms82"  "largef92" "largef87" "largef82" "smallf92"
[13] "smallf87" "smallf82" "region"
```

```
1/unique(dsrsp$prob) # See the poststratified weight for each region
```

```
[1] 10.630769 10.820513 9.850467 9.166667
```

```
svymean(~acres92, dsrsp)
```

```
      mean      SE
acres92 299778 17513
```

```
svytotal(~acres92, dsrsp)
```

```
      total      SE
acres92 922717031 53906392
```


Summary

- Ratio and regression estimation use an auxiliary variable that is highly correlated with the variable of interest to reduce the MSE of estimated population means or totals.
- The estimators in ratio and regression estimation come from models that we hope describe the data, but the randomization-theory properties of the estimators do not depend on these models.
- Ratio estimation is especially useful in cluster sampling as we shall see in the next sessions